Practical No. 1

Aim

To create and display a Pandas DataFrame using a dictionary.

Theory

Pandas is a Python library used for data manipulation and analysis. A DataFrame is a two-dimensional data structure consisting of rows and columns. A DataFrame can be created using a dictionary of lists.

Algorithm

Import the Pandas library.

Create a dictionary containing student names and marks.

Convert the dictionary into a DataFrame.

Display the DataFrame.

Stop the program.

Program

```
import pandas as pd

data = {
    'Name': ['Amit', 'Riya', 'Rahul', 'Sneha'],
    'Marks': [85, 92, 78, 88]
}

df = pd.DataFrame(data)

print("Student Details:")
print(df)
```

Output

Student Details:

Name Marks

0 Amit   85

1 Riya   92

2 Rahul   78

3 Sneha   88

Result

The Pandas DataFrame was created and displayed successfully.

Practical No. 2

Aim

To handle missing values in a Pandas DataFrame using the fillna() function.

Theory

Missing values can occur in a dataset. Pandas provides functions such as isnull() to check missing values and fillna() to replace them. The fillna(0) function replaces missing values with zero.

Algorithm

Import Pandas and NumPy libraries.

Create a DataFrame containing student marks with a missing value.

Display the original DataFrame.

Replace the missing value with 0 using fillna(0).

Display the updated DataFrame.

Stop the program.

Program

import pandas as pd

import numpy as np

```
data = {
    'Name': ['Amit', 'Riya', 'Rahul', 'Sneha'],
    'Marks': [85, np.nan, 78, 90]
}

df = pd.DataFrame(data)

print("Original Data:")
print(df)

df = df.fillna(0)

print("\nAfter Replacing Missing Value:")
print(df)
```

Output

```
Original Data:
    Name  Marks
0   Amit   85.0
1   Riya    NaN
2  Rahul   78.0
3  Sneha   90.0

After Replacing Missing Value:
    Name  Marks
0   Amit   85.0
1   Riya    0.0
```

2 Rahul 78.0

3 Sneha 90.0

Result

The missing value in the DataFrame was successfully replaced with 0 using the fillna() function.